

This lab session covers Django Models (Part 3). By the end of this session, students should have a very good understanding of how to make advanced queries using annotations.

Pre-lab Preparation:

1. The following DB models scheme

```
class Publisher(models.Model):
    name = models.CharField(max_length=200)
    location = models.CharField(max_length=300)

class Author(models.Model):
    name = models.CharField(max_length=200)
    DOB = models.DateField(null=True)

class Book(models.Model):
    title = models.CharField(max_length= 100)
    price = models.FloatField(default=0.0)
    quantity = models.IntegerField(default=1)
    pubdate = models.DateTimeField()
    rating = models.SmallIntegerField(default = 1)

    publisher = models.ForeignKey(Publisher, null=True, on_delete=models.SET_NULL)
    authors = models.ManyToManyField(Author)
```

2. Use fake data from mockaroo.com website or any other methods to fill tables above. Make sure that you handle the Book.publisher and Book.authors fields correctly.

Lab Activities:

Task 1: Create a URL '/books/lab9/task1' with any necessary HTML file and view function to list books along with percentage availability. For example, if the number of books in 350 and the quantity of a book is 7, then the percentage of availability is 2%. Use transient field for the percentage.

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Task 1 - Availability

Title	Quantity	Availability %
Continuous Delivery	1	2.22%
Continuous Delivery	1	2.22%
Networks and Security	1	2.22%
Good Son, The	1	2.22%
Vizontele	1	2.22%
Love, Wedding, Marriage	1	2.22%
Hothead (Coup de tête)	1	2.22%
Lawyer Man	1	2.22%
The Big Bus	1	2.22%
Things to Do in Denver When You're Dead	1	2.22%
Oh, God! Book II	1	2.22%
Applause (Applaus)	1	2.22%
Curious Case of Benjamin Button, The	1	2.22%
Faculty, The	1	2.22%
Undercover Man	1	2.22%
Beauty in Trouble (Kráska v nesnázích)	1	2.22%
Carousel	1	2.22%
Perfect Blue	1	2.22%
Midnight Movie	1	2.22%
Sahara	1	2.22%
We Are Young. We Are Strong.	1	2.22%
Queen and I, The (Drottningen och jag)	1	2.22%
Highly Dangerous	1	2.22%
Green Tea (Lü cha)	1	2.22%
After Alice (Eye of the Killer)	1	2.22%
Meteor	1	2.22%
Quiz Show	1	2.22%
Quiz Show	1	2.22%
Living Room of the Nation, The (Kansakunnan olohuone)	1	2.22%
A Cinderella Story: Once Upon a Song	1	2.22%
Watchmen	1	2.22%

Task 2: Create a URL '/books/lab9/task2', HTML file, and query list books that all publishers, annotated with their respective total book stock.

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Task 2 - Publishers Book Count

Publisher	Total Books
wiley.com	11
behance.net	10
ed.gov	10
soundcloud.com	15
so-net.ne.jp	0
lucadamaio.com	0

Task 3: (opposite to Task 2) Create a URL '/books/lab9/task3', HTML file, and a query to get oldest book of any of those managed by every publisher.

Task 3 - Oldest Book for Each Publisher

Publisher	Oldest Book Publish Date
wiley.com	May 3, 2026, 2:12 p.m.
behance.net	May 3, 2026, 2:12 p.m.
ed.gov	May 3, 2026, 2:12 p.m.
soundcloud.com	May 3, 2026, 2:12 p.m.

Task 4: Create a URL '/books/lab9/task4', HTML file, and query to calculate the average, min, and max price of books for each publisher.

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Task 4 - Books Price Statistics

Publisher	Average Price	Minimum Price	Maximum Price
wiley.com	37.96	0.00	150.00
behance.net	51.22	1.99	349.99
ed.gov	19.51	2.89	39.99
soundcloud.com	22.56	0.00	149.99

Task 5: Create a URL '/books/lab9/task5', HTML file, and a query to list publishers with a count of highly rated books, with their names and quantity.

Task 5 - Highly Rated Books Count

Publisher	Highly Rated Books
wiley.com	0
behance.net	0
ed.gov	0
soundcloud.com	0

Task 6: Create a URL '/books/lab9/task6', HTML file, and a query to count of books for each publisher, filter only books with price greater than 50 and quantity less than 5 and greater than or equal to 1.

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Task 6 - Filtered Books Count

Publisher	Book Count
wiley.com	2
behance.net	2
ed.gov	0
soundcloud.com	2
iso-net ne in	0